Casimir Falsification Criteria

Quantitative predictions in the transition region $d \to L_\xi$

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Preamble

This document formulates an experimentally testable falsification criterion for FFGFT in the Casimir region, in direct response to the methodological request from P. Austin (mail of 9 May 2026, IPI discussion): *"What scale-dependence or residual differs from the standard $1/d^4$ form, given that the ordinary $1/d^4$ form is asymptotically recovered?"*

The answer is quantitative and lies in an experimentally accessible range where currently no precision measurements exist.

1 Starting point: what FFGFT confirms

FFGFT confirms standard Casimir theory throughout the entire range measured to date. Specifically:

- **Decca et al. (170–300 nm, 0.5 % accuracy):** no measurable FFGFT correction expected.
- **Bressi et al. (0.5–3 μm):** no measurable FFGFT correction expected.
- **Lamoreaux (0.6–6 μm):** no measurable FFGFT correction expected.

In the sub-μm to few-μm range, the FFGFT correction falls off as the fourth power of (d/L_ξ) and is therefore invisible (see §2).

The falsification criterion lies in the as-yet-unmeasured range $d \sim L_\xi \approx 100\,\mu$m, where current torsional balance experiments (Iannuzzi group) and planned measurements (Dartmouth, ILL Grenoble) are just beginning to deliver precise data.

2 The FFGFT transition function

2.1 Asymptotic form (established)

From Doc. 091, the Casimir energy density in the classical region $d \ll L_\xi$ is:

$$|\rho_{\text{Casimir}}(d)| \;=\; \frac{\pi^2}{240\,\xi_0}\,\rho_{\text{CMB}}\left(\frac{L_\xi}{d}\right)^4 \;=\; \frac{\pi^2\hbar c}{240\,d^4}, \tag{1}$$

with $L_\xi = (\xi_0\hbar c/\rho_{\text{CMB}})^{1/4} \approx 100.24\,\mu$m and $\xi_0 = 4/30000$.

This is the standard formula – exactly what Decca, Bressi, and Lamoreaux measured, within their accuracy.

2.2 Transition function for $d \to L_\xi$

In FFGFT, L_ξ is not an arbitrary scale, but the vacuum length scale at which the Casimir energy density between two plates coincides with the CMB energy density:

$$\rho_{\text{Casimir}}(d = L_\xi) \;=\; \frac{\pi^2}{240}\,\rho_{\text{CMB}}\,\xi_0^{-1}. \tag{2}$$

For $d > L_\xi$, the Casimir mechanism (truncation of vacuum modes between the plates) can no longer operate, since the wavelengths of the relevant modes would exceed the plate separation.

This implies a **geometrically motivated transition function**:

$$\rho^{\text{FFGFT}}_{\text{Casimir}}(d) \;=\; \frac{\pi^2\hbar c}{240\,d^4}\cdot\mathcal{F}\left(\frac{d}{L_\xi}\right), \tag{3}$$

with the dimensionless transition function

$$\mathcal{F}(x) \;=\; \frac{1}{1+x^4}, \qquad x = d/L_\xi. \tag{4}$$

2.3 Justification of the form $1/(1+x^4)$

The choice of this transition function is not arbitrary:

- **Asymptote** $x \to 0$ **(i.e.,** $d \ll L_\xi$**):** $\mathcal{F} \to 1$, standard $1/d^4$ is reproduced exactly.
- **Asymptote** $x \to \infty$ **(i.e.,** $d \gg L_\xi$**):** $\mathcal{F} \to 1/x^4$, hence $\rho_{\text{Casimir}} \to \rho_{\text{CMB}}$ constant – the vacuum between the plates is no longer distinguishable from the CMB.
- **Transition point** $x = 1$ **(**$d = L_\xi$**):** $\mathcal{F}(1) = 1/2$, exactly the crossover between the two regimes.
- **Power** 4**:** follows from the four-dimensionality of the energy-density volume $\hbar c/L^4$ and is consistent with the CMB bridge formula from Doc. 091.

3 Quantitative predictions

From (4) the relative deviation from the standard formula follows:

$$\Delta(d) \;\equiv\; \frac{\rho^{\text{FFGFT}}_{\text{Casimir}}(d) - \rho^{\text{std}}_{\text{Casimir}}(d)}{\rho^{\text{std}}_{\text{Casimir}}(d)} \;=\; \frac{1}{1+(d/L_\xi)^4} - 1 \;=\; -\frac{(d/L_\xi)^4}{1+(d/L_\xi)^4}. \tag{5}$$

With $L_\xi = 100\,\mu$m, the following numbers result:

d	$(d/L_\xi)^4$	$\lvert\Delta\rvert$	**Measurable?**
100 nm	10^{-12}	$10^{-10}\,\%$	no
1 μm	10^{-8}	$10^{-6}\,\%$	no
3 μm	$8.1\cdot10^{-7}$	$\sim 10^{-4}\,\%$	no
10 μm	10^{-4}	$\approx 0.01\,\%$	marginal
30 μm	$8.1\cdot10^{-3}$	$\approx 0.80\,\%$	**yes**
50 μm	0.063	$\approx 5.9\,\%$	**yes**
70 μm	0.24	$\approx 19.3\,\%$	**yes**
100 μm	1.0	$50\,\%$	**yes**

3.1 Three falsification regimes

Regime A: $d < 10\,\mu$m – confirmation range. FFGFT predicts $|\Delta| < 0.01\,\%$. All existing precision measurements (Decca: 170–300 nm; Bressi: 0.5–3 μm; Lamoreaux: up to 6 μm) lie in this range. *FFGFT does not contradict any of these experiments.*

Regime B: $10\,\mu$m $< d < 100\,\mu$m – falsification range. FFGFT predicts a growing, monotonically increasing deviation from $0.01\,\%$ (at $d = 10\,\mu$m) to $50\,\%$ (at $d = L_\xi$). Currently no precision measurements exist in this range, but the required sensitivity ($\sim 1\,\%$ at $d = 30$–$50\,\mu$m) is achievable with torsional balance setups (Iannuzzi group, Dartmouth, ILL Grenoble).

Regime C: $d \to L_\xi$ – prediction point. At exactly $d = L_\xi$, $\rho_{\text{Casimir}} = (1/2)\rho^{\text{std}}_{\text{Casimir}}$ should be measured, with L_ξ fixed by Doc. 091 to $100.24\,\mu$m. This is the *sharpest* test: a single measurement point at $100\,\mu$m with $\sim 5\,\%$ accuracy fixes or refutes the FFGFT prediction.

4 Specific falsification criteria

FFGFT is **refuted** by any of the following observations:

1. **Sub-μm range:** A Casimir measurement at $d < 1\,\mu$m showing a deviation $|\Delta| > 0.1\,\%$ from standard $1/d^4$. *Reason:* FFGFT predicts $|\Delta| < 10^{-6}\,\%$ here.
2. **Transition range:** A precision measurement in the range 30–$70\,\mu$m showing *no* systematic negative deviation of the size predicted by (5). *Reason:* FFGFT predicts $|\Delta|$ between $1\,\%$ and $20\,\%$ here, with definite sign (Casimir force weaker than $1/d^4$).
3. **Prediction point:** A measurement at $d = 100\,\mu$m that does not give $\rho_{\text{Casimir}}(L_\xi) \approx \frac{1}{2}\,\rho^{\text{std}}_{\text{Casimir}}(L_\xi)$ (within measurement accuracy), or yields L_ξ significantly different from the value $L_\xi^{\text{T0}} = 100.24\,\mu$m calculated from ρ_{CMB}. *Reason:* The bridge to the CMB (Doc. 091, eq. **??**) would be broken.
4. **Functional form:** A systematic deviation of the measured transition function from the form $\mathcal{F}(d/L_\xi) = 1/(1 + (d/L_\xi)^4)$ outside measurement uncertainty.

Each of these criteria is sufficient for refutation of the FFGFT Casimir prediction. They are mutually *independent*, so several experiments can test the theory in parallel or sequentially.

5 What FFGFT does NOT predict

For methodological clarity:

- FFGFT does *not* predict an additional Yukawa-type force $\sim e^{-d/\lambda}$. Modifications from extra dimensions (Randall-Sundrum, ADD) produce such corrections; FFGFT does not. A measured Yukawa term is neither evidence for nor against FFGFT.
- FFGFT does *not* predict modifications to the thermal Casimir correction (Drude vs. plasma model debate). The FFGFT prediction is orthogonal to that debate and compatible with both models.

- FFGFT does *not* predict modifications to geometric factors (sphere-plate vs. plate-plate, Proximity Force Approximation). The correction (4) is universal in d/L_ξ and should appear identically in all geometries.

6 Summary

FFGFT makes a quantitative, experimentally testable prediction in the Casimir region:

$$\boxed{\rho_{\text{Casimir}}^{\text{FFGFT}}(d) \;=\; \rho_{\text{Casimir}}^{\text{std}}(d) \cdot \frac{1}{1+(d/L_\xi)^4}, \qquad L_\xi = 100.24\,\mu\text{m}.}$$

The prediction:

- is compatible with all existing data in the measured range ($d < 10\,\mu$m, confirmation regime, $|\Delta| < 0.01\,\%$);
- delivers a measurable, monotonically increasing deviation (0.8–$50\,\%$) in the range 30–$100\,\mu$m;
- is experimentally accessible with current torsional balance technology;
- contains no free parameter – L_ξ is fixed numerically through Doc. 091 by ρ_{CMB};
- is falsifiable on four independent criteria (sub-μm, transition range, prediction point, functional form).

Methodological note. This prediction is FFGFT-internal. It follows from the ξ-vacuum geometry and the Casimir-CMB bridge formula (Doc. 091), not from a topological partition or a recursive survival structure. Whether other IPI strands can reproduce the same prediction in their own language remains open and is not a precondition for the validity of the statement.

Status. The prediction stands available for experimental test. Should existing or upcoming torsional balance data in the 30–$100\,\mu$m range become available, they are to be compared directly with (4).